

Pratham Mehta

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Education

University of Virginia

Charlottesville, VA | **Expected Graduation: December 2025**

Computer Science (Bachelor of Science), Applied Mathematics (Minor)

- Relevant Coursework: Data Structures and Algorithms, Software Engineering, Autonomous Vehicles, Artificial Intelligence, Machine Learning in Image Analysis, Cybersecurity, Probability, Linear Algebra, Data Analysis, Statistics, Calculus 3
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Skills and Certifications

Languages: Python, Java, SQL, TypeScript, JavaScript, HTML, React, Django, R

Frameworks and Libraries: ROS, React, Django, FastAPI, PyTorch, TensorFlow, pandas, numpy, Optuna, Keras Tuner, Matplotlib

Tools and Databases: Git, Bash, MongoDB, PostgreSQL, Linux, Docker, AWS, Azure, CI/CD, unit testing, API design, SageMaker

Certifications: AWS Cloud Practitioner, AWS AI Practitioner, Python Certified Associate

Experience

Software and AI Intern, **Reveal Global Consulting (US Census)**

Fulton, MD | **June 2025 - August 2025**

- Enhancing satellite-based construction tracking by enhancing a production-grade computer vision pipeline (~2M images/month)
- Achieved a 7% accuracy gain (85% to 92%) in building-type classification by designing and implementing an ensemble pipeline combining HRNet for feature extraction with XGBoost for classification
- Reduced model evaluation time by 70% through automating evaluation of 7 concurrent models with a full-stack Django-based dashboard for bottleneck detection, frequency distributions, and confusion matrix generation, reducing manual error rate by 13%

ML Research Assistant, **UVA Physics (Fermilab)**

Charlottesville, VA | **October 2024 - May 2025**

- Simulated and predicted quantum particle track reconstruction using TensorFlow, Monte Carlo methods, and ROOT files
- Achieved 12% increase in reconstruction accuracy and ~50% reduction in training time by integrating Optuna-based hyperparameter optimization into reproducible model training pipeline

Data Science Intern, **Hitachi Vantara Federal**

Reston, VA | **June 2024 - August 2024**

- Built scalable legal document recommender using fine-tuned BERT embeddings and Pinecone vector database, improving top-3 recommendation accuracy by 120% over TF-IDF baseline and cutting query time to < 90s across a corpus of 1M+ entries
- Uncovered trends between zip-code and loan amounts by ingesting, processing, and visualizing >7.2 million SBA records using SQL, PostgreSQL and Pentaho+

Machine Learning Researcher, **Biocomplexity Institute**

Charlottesville, VA | **October 2023 - February 2024**

- Achieved ~80% accuracy in predicting commodity flow with a spatial-temporal graph neural network (ST-GNN) in PyTorch, leveraging 50+ research papers to refine model structure and enhance predictive insights
 - Improved prediction accuracy by up to 20% by testing Geometric Temporal Package against traditional models, identifying optimal configurations for commodity forecasting
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Software Engineering Projects

Autonomous Racing Algorithm | Python, ROS, Git, RViz

- Fine-tuned PID control system to maintain consistent lateral distance from walls, reducing deviation to under 2 cm and improving stability at speeds up to 4 m/s
- Mapped 2D representation of the racetrack and localized racecar using Adaptive Monte Carlo Localization (AMCL), achieving positional accuracy within 1.5 cm when testing and visualizing with RViz
- Improved race time by 24% by implementing waypoint logging on a pre-calculated reference trajectory, boosting velocity from 4 m/s to 6 m/s without deviation

Fine-Tuned Financial LLM + Synthetic Data Generator | Python, OpenAI, LangChain, Llama, Twilio, SerpAPI, Git, PostgreSQL

- Enabled fine-tuning on wealth management chatbot with recursive pipeline with multiple generative AI agents (professor, student, moderator) producing 300+ synthetic conversations with topic abstraction through YAML configurations
- Prototyped financial education chatbot by leveraging Twilio for automated SMS delivery to > 50 users and davinci-002 to generate educational, context-aware responses
- Automated acquisition of 150+ cryptocurrency research papers using Python, establishing a foundational dataset for fine-tuning Llama 3.1 on cryptocurrency topics

Depression Screening Chatbot | Python, OpenAI, Git, MongoDB, Express, React, Node.js

- Developed depression-screening chatbot for increased accessibility and privacy under Professor Panagiotis Apostolellis
- Enhanced chatbot's user engagement and acceptance rating by integrating support for diverse ethnicities, customizing design to reflect culturally relevant colors and names based on user feedback from Qualtrics surveys
- Integrated MongoDB for persistent message storage, enabling session continuity and improved longitudinal screening